

MD RAKIBUL HASAN

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U.S. Permanent Resident – authorized to work without sponsorship

Summary

Machine learning professional with 7+ years of experience building predictive models and data-driven optimization solutions on 12TB+ datasets across a 53M-subscriber network. Developed and applied ML methods for churn prediction, time-series forecasting, customer segmentation, and campaign optimization, driving 11% regional revenue growth and 16% improvement in campaign efficiency. PhD candidate in Computer Science at NJIT, with 6 peer-reviewed publications on reinforcement learning and adaptive optimization, and a track record of building LLM systems and translating large-scale data into deployable ML solutions with measurable business impact.

Tech stack: Python, SQL, Scikit-learn, PyTorch, Hive/Hadoop/Spark, Databricks, Docker, Power BI, Git.

Education

PhD in Computer Science, New Jersey Institute of Technology (NJIT), USA December 2026
GPA: 3.91/4.0

MBA, Institute of Business Administration, University of Dhaka, Bangladesh 2021
GPA: 3.82/4.0 (*Valedictorian*)

BS in Electrical & Electronic Engineering, BUET, Bangladesh 2013

Experience

PhD Researcher (ML, Database) 09/2021 – Present
NJIT, Newark, NJ, USA

- Reduced SQL query execution time by **30%** on dynamic workloads by developing a **Dueling NoisyNet-DQN** tuner for adaptive database index selection, outperforming existing learning-based baselines.
- Lowered total query time and index transition overhead by building a **Bayesian Q-learning framework** for online index tuning, achieving **21× less transition cost** than Multi-Armed Bandit and RL-based advisors via Thompson Sampling.
- Accelerated graph pattern query evaluation by **2.7×** by designing a **summary graph-based materialized view** framework that prunes search space while reducing storage requirements on large real-world graphs.

Manager (ML & DS), Telecom Market Strategy & Planning 10/2020 – 08/2021
Robi Axiata (53M+ subscribers), Bangladesh

- Improved regional campaign efficiency by **16%** by building **segmentation models** across geographic, demographic, revenue, and cost dimensions to prioritize market investments.
- Developed **dynamic time-series forecasting** models for revenue chains achieving **95% forecast accuracy**, enabling strategic investment decisions at regional and national levels.
- Conducted **customer churn analysis** combining CDP data and network traffic insights; recommended targeted retention strategies that delivered an **11% regional revenue boost**.
- Designed and deployed an **ML-driven marketing campaign** for OTT platform (Binge), increasing new user acquisitions by **21%** and improving campaign efficiency by **16%** through personalized promotions.
- Led monthly/quarterly performance reviews with variance analysis, resulting in **12% improvement in regional KPIs**.

Specialist, Network Planning (ML) 09/2016 – 09/2020
Robi Axiata Limited, Bangladesh

- Enabled proactive capacity planning that prevented up to 12% daily traffic loss during the COVID-19 demand shift, forecasting region-to-region user mobility with Markov/HMM models and per-region traffic load with SARIMA time-series models.
- Accelerated 4.5G rollout by training clustering models to prioritize 7,000 high-potential LTE sites across 64 districts, directing build-out toward the regions with the strongest projected demand.
- Reduced radio-controller (RNC) utilization 15% by modeling the parameters that most drive RNC/BSC processor load with a neural network, then rehomeing sites on the data-driven recommendations.
- Led RF integration of 4,500 sites during the Robi-Airtel merger, the largest merger in Bangladesh and directed 23 engineers and cut customer complaints 29% within three days of rehomeing.

Research Intern (ML & DS) 09/2019 – 12/2019
Fujitsu Research Institute, Tokyo, Japan

- Improved autonomous vehicle braking system response by analyzing time-series car sensor data using **LSTM**, reducing false-positive braking events in simulation testing.
- Increased convenience-store daily sales by **2.1%** and customer engagement by **3.6%** by mining POS data with Gradient Boosting, Eclat, and Apriori algorithms.
- Predicted automotive manufacturer monthly demand at **95% confidence** using LSTM, SARIMA, and Prophet on his-

torical sales and macroeconomic data.

RF Optimization Engineer

01/2014 – 08/2016

Huawei Technologies, Bangladesh

- Enabled real-time monitoring of **3,800+ NodeBs** by implementing **enhanced Markov algorithm** for user mobility prediction and **MapReduce** pipelines for network KPI/KQI analytics on a Hadoop big data platform.
- Automated targeted promotional campaigns across millions of subscribers by developing **scalable ML classification models** on Hadoop to segment customer interests, replacing manual campaign selection.
- Transformed managed-service projects from monthly penalty to reward by leading **7 optimization teams** covering **1,450+ NodeBs** and optimizing KPIs (RRC failure, RAB assignment, CS/PS drop, congestion).

Projects

- Database Q&A Chatbot (RAG):** Built a retrieval-augmented (RAG) Q&A chatbot over a 631-page technical corpus, delivering source-cited, hallucination-free answers on 5 of 6 benchmark questions by combining ChromaDB semantic retrieval, a vision LLM that recovered 100+ diagram/math slides (~16% of the corpus), and a 6-clause grounding prompt at temperature 0.1.
- LLM Fine-Tuning Pipeline:** Fine-tuned Llama-2 7B for text-to-SQL generation using LoRA and QLoRA with Hugging Face PEFT. Deployed inference endpoint on AWS SageMaker with 4-bit quantization.
- Multi-Signal AI-Text Detection API:** Designed and deployed a multi-signal AI-text detection API by integrating a Llama-3.3-70B holistic evaluator, stylometrics, and lexical density into a Flask ensemble, minimizing false positives via asymmetric thresholds and ensuring traceable verdicts through structured audit logs.
- Fine-Tuned vs Zero-Shot LLM Benchmark:** Benchmarked fine-tuned DistilBERT against zero-shot Llama-3.3-70B for text classification, using confusion-matrix error analysis on sparse domain vocabulary to identify a 20-point zero-shot accuracy advantage (90% vs 70%) and establish a clear ML strategy for Network question datasets.
- LLM Rule Authoring + Deterministic Engine:** Developed a hallucination-free, human-in-the-loop insurance claim-edit pipeline by coupling LLM-drafted medical rules with a deterministic Pandas execution engine, delivering an auditable Streamlit application for processing PTP, MUE, and demographic claims.
- Customer Lifetime Value Model:** Built probabilistic CLV prediction using BG/NBD and Gamma-Gamma models for marketing budget allocation.

Technical Skills

- Languages:** Python, C++, SQL, MATLAB
- ML & Data Science (DS):** Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, SciPy; Regression, Classification, Clustering, Time-Series Forecasting, Anomaly Detection, Reinforcement Learning, Deep Learning (LSTM, CNN, DQN), GenAI, LLM, Agentic AI
- Big Data & Databases:** Hive SQL, Hadoop/HDFS, Apache Spark, Databricks, MySQL, PostgreSQL, IBM Db2
- Visualization & BI:** Power BI, Tableau, Excel VBA
- Tools & Methods:** Git, SPSS, SAS, Statistical Modeling, A/B Testing, ETL Pipelines, Feature engineering pipeline, Batch inference, Docker

Publications

- “Multi-Objective Reinforcement Learning for Index Selection in Hybrid Transactional/Analytical Workloads” — **ICDE (In review)**
- “Database Tuning via Distributional Reinforcement Learning for Hybrid Transactional/Analytical Processing Workloads” — **DaWaK 2026**
- “Adaptive Online Index Selection with a Noisy Dueling Deep Q-Network” — **WISE 2025**
- “A Bayesian Reinforcement Learning Framework for Online Index Tuning” — **DaWaK 2025**
- “Scalable Optimization of Graph Pattern Queries Using Summary Graphs” — **WISE 2024**
- “LiteSelect: A Lightweight Adaptive Learning Algorithm for Online Index Selection” — **DaWaK 2024**
- “Performance of successive interference cancellation technique in IEEE 802.11 based WLANs” — **ICIEV 2013**

Certifications

- Scalable Machine Learning on Big Data using **Apache Spark** — Coursera, 2020
- Data Science on **Google Cloud Platform: Predictive Analytics** — Coursera, 2020
- IBM AI Engineering Specialization** — Coursera

Awards & Recognition

- Gold Medal & Valedictorian**, Executive MBA, IBA, University of Dhaka (highest CGPA, 2021)
- Merger Hero 2017**, Robi Axiata — for leading RF integration during Robi-Airtel merger preserving baseline KPIs
- Q2'2019 Technology Champion**, Robi Axiata — for managing network capacity during festival traffic surge